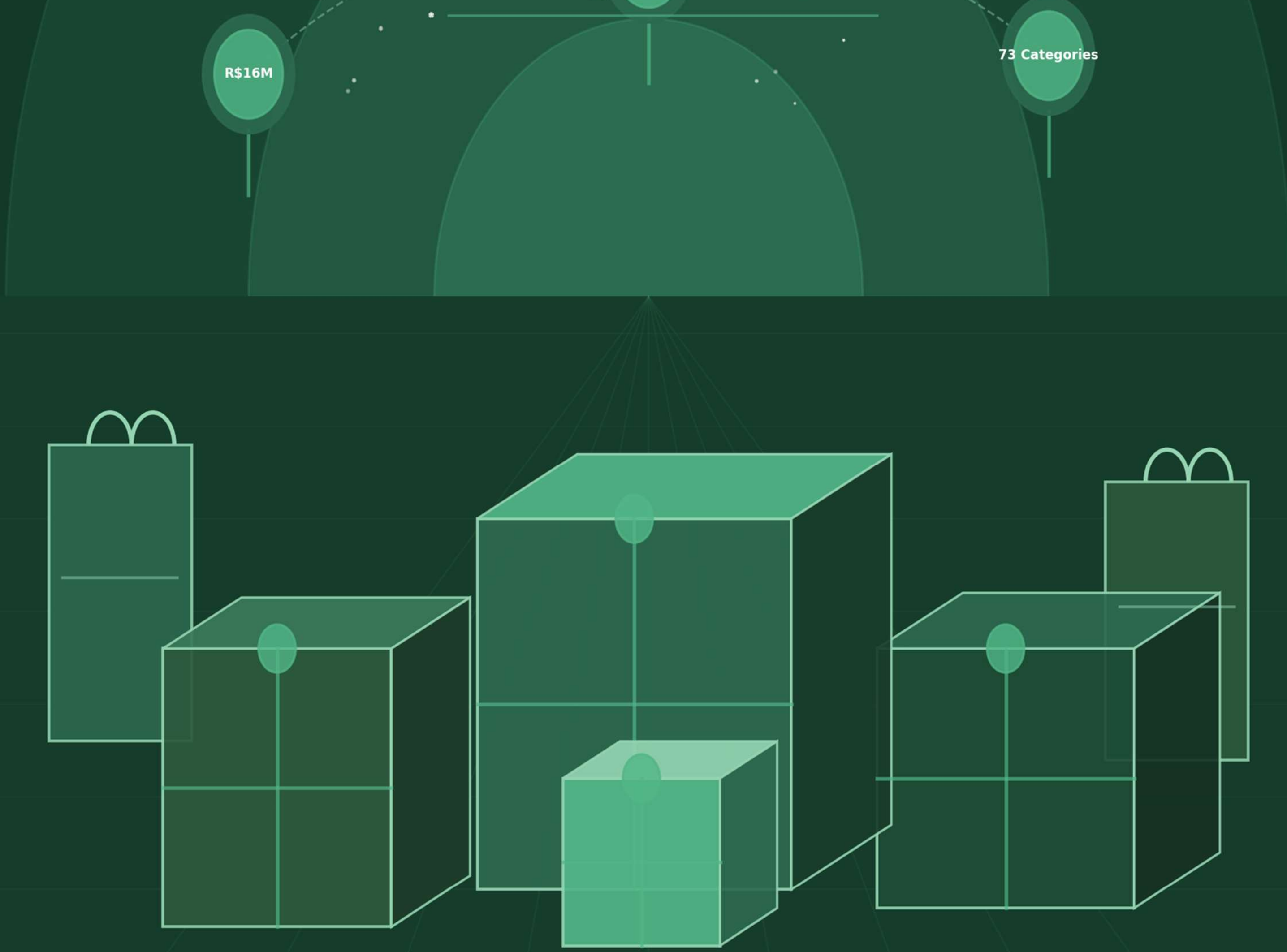
Customer Intelligence
& Market Insights

A Data Mining Analysis of the Brazilian E-Commerce Dataset

AUTHOR

Digen Naik



Customer Segmentation • K-Means • Decision Tree • Apriori Rules

Executive Summary

This report presents a comprehensive data mining analysis of the Olist Brazilian E-Commerce dataset, covering 112,650 orders placed between October 2016 and August 2018. Three business questions guided the analysis: who are the customers, what do they buy together, and what drives high-value purchases.

Two analytical pipelines were developed. The first combined unsupervised **K-Means clustering** with a supervised learning and **Decision Tree classifier** to segment customers into three behaviorally distinct groups Low-Value Casual shoppers (77.7%), Installment Buyers (19.0%), and High-Value Shoppers (3.3%). The Decision Tree achieved **99% accuracy** on a held-out test set of 9,648 customers.

The second pipeline applied **Apriori association rule mining** to multi-basket orders, uncovering six co-purchase patterns. The strongest Health & Beauty → Perfumery carried a lift of 3.8, meaning Health & Beauty buyers are 3.8× more likely to also purchase perfumery products.

Together, these findings support a set of concrete prescriptive recommendations: loyalty programs for High-Value Shoppers, re-engagement campaigns for Casual buyers, upsell strategies for Installment Buyers, and product recommendation engines powered by the discovered association rules.

Table Of Contents

1. Introduction

1.1 Dataset Overview

The Olist Brazilian E-Commerce dataset is a publicly available collection of anonymized commercial transactions from the Olist marketplace one of Brazil's largest e-commerce platforms. The dataset spans October 2016 to August 2018, spread across nine relational CSV files covering customers, orders, payments, products, sellers, reviews, and geolocation data.

The dataset is rich in behavioral and transactional detail, making it well-suited for customer segmentation and market basket analysis. Key characteristics include:

Metric	Value
Total Orders	112,650
Total Revenue	R$16.01 Million
Average Order Value	R$154.10
Average Freight Value	R$19.99
Total Freight Revenue	R$2.25 Million
Distinct Product Categories	73
Brazilian States Covered	27
Single-Order Customers	96.72%
Peak Order Month	November

1.2 Research Questions

- **Q1)** Who are our customers? Can customers be segmented by spending behavior, frequency, and payment patterns?
- **Q2)** What do customers buy together? Are there product combinations appearing more often than random chance predicts?
- **Q3)** What drives high-value purchases? Which categories and payment behaviors are associated with higher order values?

1.3 Analytical Methods

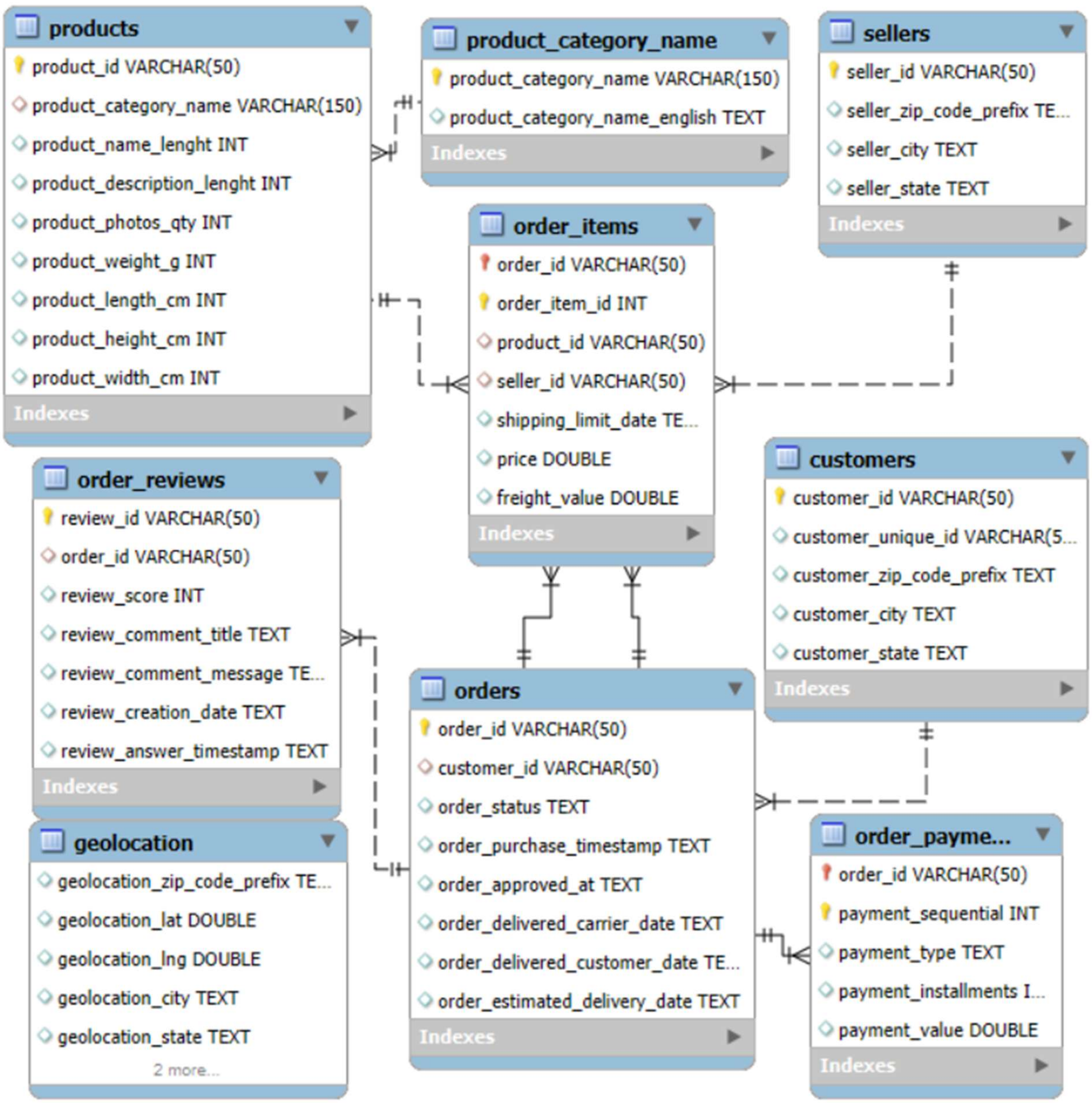
Method	Purpose	Business Question
K-Means Clustering	Unsupervised customer segmentation	Q1
Decision Tree Classification	Scalable real-time segment prediction	Q1
Apriori Association Rules	Co-purchase pattern discovery	Q2, Q3

2. Data Preparation

2.1 Database Construction in MySQL Workbench

The nine raw CSV files were imported into MySQL Workbench to create a structured relational database. Primary and foreign key relationships were defined across all tables to enable efficient multi-table querying:

- orders.order_id → primary key linked to order_items, order_payments, order_reviews
- orders.customer_id → foreign key linking to customers
- order_items.product_id → foreign key linking to products
- products.product_category_name → linked to product_category_name_translation for English labels



This relational structure allowed complex joins to be performed at the database layer rather than in Python, reducing data redundancy and ensuring referential integrity before any machine learning was applied.

2.2 Data Cleaning & Filtering

Several preprocessing decisions were made to ensure data quality:

- **Order status filter:** Only 'delivered' orders were retained. Cancelled, unavailable, and in-transit orders were excluded.
- **Null handling:** Rows with null payment value or order purchase timestamp were removed (<0.5% of data). Rows with null freight value were replaced with zero because there were some promotional coupons on free delivery for small period.
- **Data type casting:** Timestamp columns were cast from string to datetime for recency calculations.
- **Duplicate orders:** Each customer appeared only once after grouping by customer id.

2.3 Feature Engineering

A SQL query was written to extract customer-level behavioral features from the relational database and export the result as a CSV file for Python:

```
SELECT customer_id,
    COUNT(DISTINCT order_id)        AS frequency,
    SUM(payment_value)              AS monetary,
    MAX(order_purchase_timestamp)   AS last_purchase,
    AVG(payment_installments)       AS avg_installments,
    AVG(freight_value)              AS avg_freight,
    COUNT(DISTINCT product_id)      AS unique_products,
    customer_state
FROM orders
JOIN order_payments ON orders.order_id = order_payments.order_id
JOIN customers      ON orders.customer_id = customers.customer_id
WHERE order_status = 'delivered'
GROUP BY customer_id, customer_state;
```

The resulting feature set captured six dimensions of customer behavior:

Feature	Description	Business Rationale
Recency	Days since last purchase	Identifies active vs. lapsed customers
Frequency	Number of distinct orders placed	Distinguishes loyal from one-time buyers
Monetary	Total payment value across all orders	Measures direct revenue contribution
Avg Installments	Average number of payment installments	Reflects payment behavior and commitment
Avg Freight	Average freight value paid	Proxy for product size or geographic distance
Unique Products	Count of distinct product IDs purchased	Measures breadth of category engagement

Recency was calculated in Python as the difference in days between each customer's last_purchase date and the maximum date in the dataset, ensuring a consistent reference point

2.4 Feature Scaling

K-Means clustering is sensitive to the scale of input features, as it uses Euclidean distance to measure similarity. Without scaling, high-magnitude features such as monetary value (ranging from R$10 to R$10,000+) would dominate the clustering over lower-magnitude features like frequency (typically 1–5). All six features were standardized using StandardScaler from scikit-learn, transforming each feature to have a mean of zero and a standard deviation of one.

2.5 Data Preparation for Association Rules

A separate data pipeline was constructed for the Apriori analysis. The starting point was a descriptive finding: **96.72% of all orders contained only a single product category**. Association rule mining requires at least two items per transaction to detect co-purchase patterns. Therefore, orders were filtered to retain only **multi-basket orders** containing two or more distinct product categories resulting in approximately **2,880 usable transactions** (2.88% of all orders).

The preparation steps were:

1. Import order_items and products tables from the database
2. Join with product_category_name_translation to obtain English category names
3. Filter to multi-basket orders (orders with 2+ distinct categories)
4. Apply transaction encoding: each row represents one order, each column represents a product category, with binary values (1 = purchased, 0 = not purchased)

Following is the SQL query:

```
SELECT
  o.order_id,
  t.product_category_name_english
FROM order_items oi
JOIN orders o ON oi.order_id = o.order_id
JOIN products p ON oi.product_id = p.product_id
LEFT JOIN product_category_name t
ON p.product_category_name = t.product_category_name
WHERE t.product_category_name_english IS NOT NULL
ORDER BY o.order_id
INTO OUTFILE 'C:/ProgramData/MySQL/MySQL Server 8.0/Uploads/association.csv'
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n';
```

3. Descriptive Analytics

Before building models, descriptive analysis was conducted to establish a baseline understanding of the dataset, validate feature selection, and identify key behavioral patterns. The Power BI dashboard below summarizes core KPIs, category-level revenue, seasonal trends, and installment behavior.

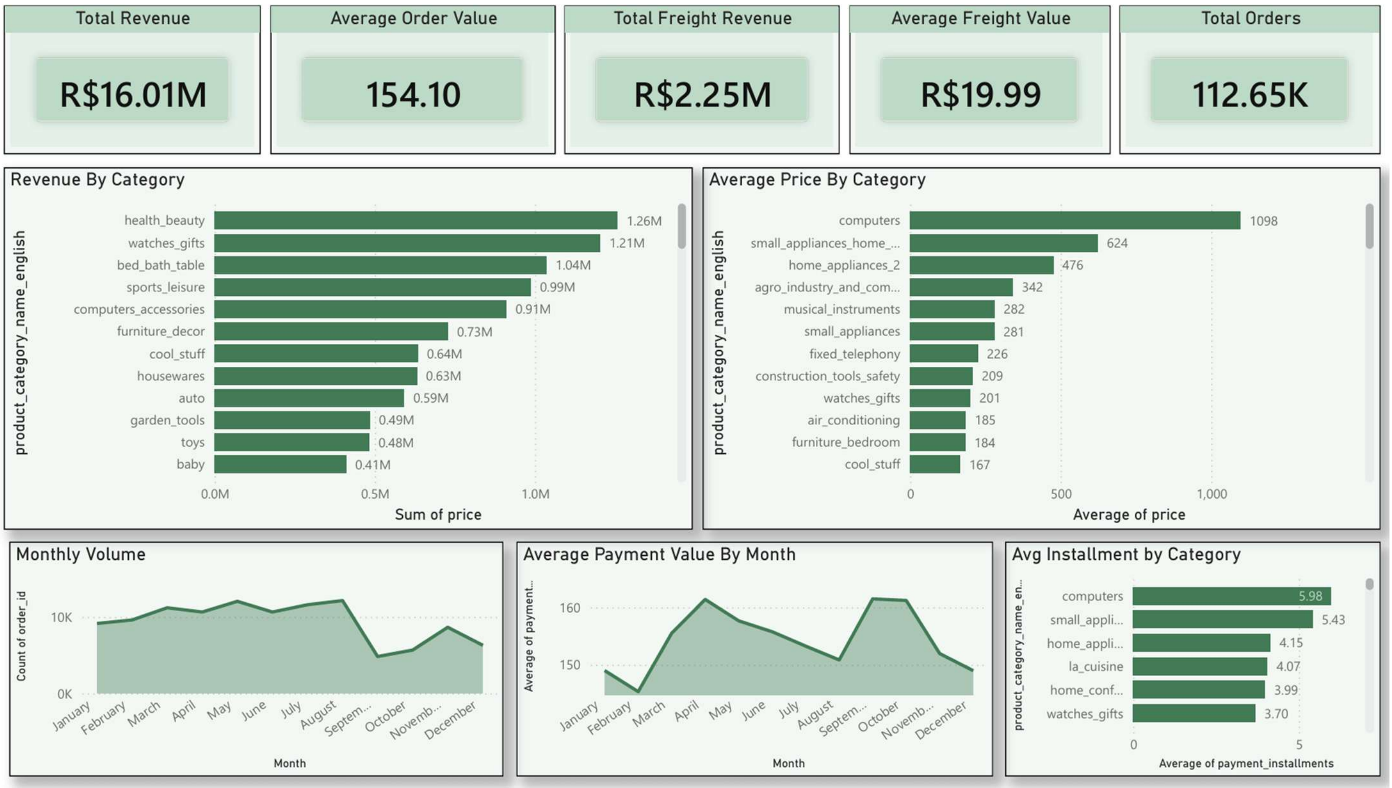


Figure 1: Power BI Descriptive Analytics Dashboard

3.1 Revenue by Product Category

Health & Beauty was the highest-revenue category at R$1.26M, followed by Watches & Gifts (R$1.21M) and Bed & Bath Table (R$1.04M). Computers carried an average price of R$1,098 nearly 5× the dataset average of R$154.10 revealing two distinct buyer archetypes: high-frequency moderate-spend buyers and low-frequency high-spend buyers.

3.2 Payment Installment Behavior

Payment installment usage varied significantly across product categories. Computers averaged 5.98 installments per order the highest of any category while Baby products averaged only 2.88. This wide variation (nearly a 2x difference) confirmed that installment behavior is a meaningful signal of both product type and buyer financial behavior, justifying its inclusion as a clustering feature.

3.3 Central Business Problem

A critical observation from the descriptive analysis is that average monthly payment value increased in September and October which decreased monthly total orders. This shows customers quickly switches to different sites for cheaper price. As the average payment value decreased monthly orders increased.

4. K-Means Clustering

4.1 Method Overview

K-Means is an unsupervised machine learning algorithm that partitions a dataset into K clusters by minimizing the within-cluster sum of squared distances (inertia) between each data point and its cluster centroid. The algorithm iterates between two steps: assigning each observation to the nearest centroid, and recalculating centroids as the mean of all assigned observations. This process continues until centroids stabilize.

Before running algorithm, we split the data in 90/10 split where 90% data will be used for training and 10% will be the unseen data which will be used for testing Decision Tree model on unseen data.

4.2 Selecting the Optimal K Elbow & Silhouette

The optimal value of K was determined using two complementary methods first is Elbow method and other one is Silhouette Score:

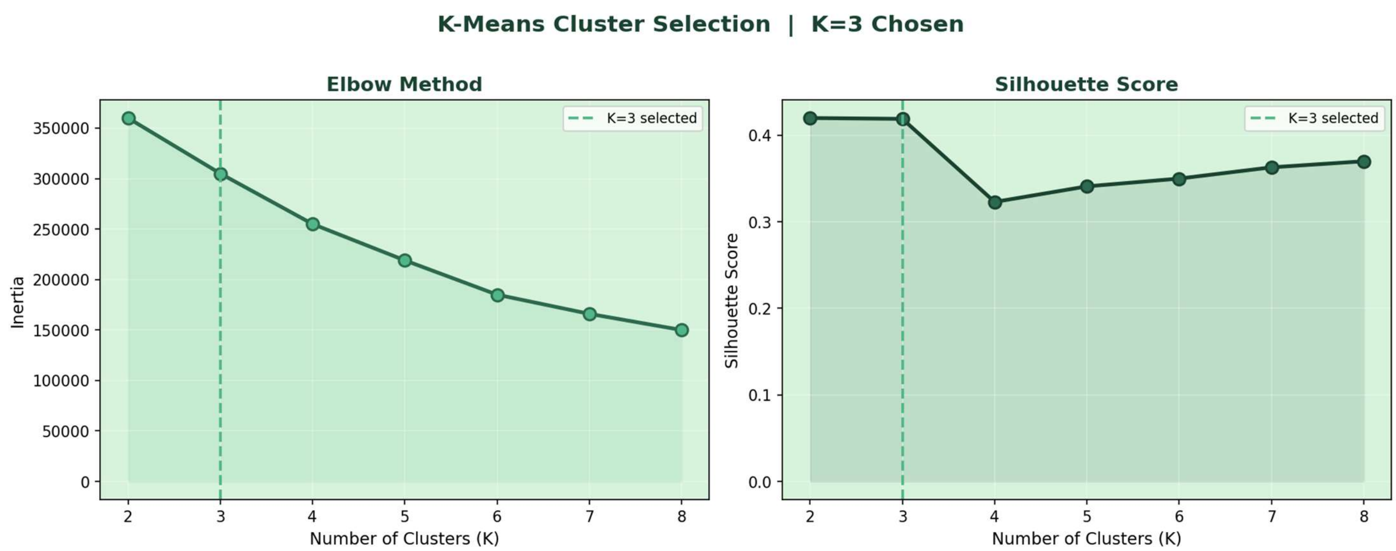


Figure 2: Elbow Method & Silhouette Score - K=3 Selected (dashed vertical line)

- Elbow Method: Inertia was plotted for K=2 to K=8. Here there is no elbow or there is no point where the change in inertia is decreased on huge scale. So, we cannot conclude anything from Elbow method.

K	Silhouette Score	Decision
2	0.420	Valid but only binary split
3	0.419	Selected 3 interpretable segments
4	0.323	Ruled out sharp quality drop
5–8	0.341–0.370	Not selected

- Silhouette Score: The silhouette score measures how similar each point is to its own cluster compared to other clusters, ranging from -1 (incorrect assignment) to +1 (perfect separation). Scores were:
  The scores for K=2 and K=3 are nearly identical (0.420 vs 0.419), indicating both are statistically valid choices. **K=3 was selected** because it produces three distinct, business-interpretable segments rather than a single binary split. K=4 was ruled out as the silhouette score dropped sharply to 0.323, indicating weaker cluster separation.

4.3 Cluster Profiles

After fitting K-Means with K=3 on the scaled RFM dataset, cluster centroids were inverse-transformed to their original scale for interpretation. Three distinct customer segments emerged Low-Value casual Customers, Installment Buyers and High value Shoppers. These were identified by using the following output from the K-Means algorithm

Segment	Count	% Total	Avg Spend	Avg Installments	Avg Freight	Unique Products
Low-Value Casual	67,469	77.7%	R$134	1.80	R$16.88	0.98
Installment Buyers	16,509	19.0%	R$384	7.25	R$32.56	0.99
High-Value Shoppers	2,851	3.3%	R$817	4.05	R$18.38	2.16

4.4 Segment Interpretation

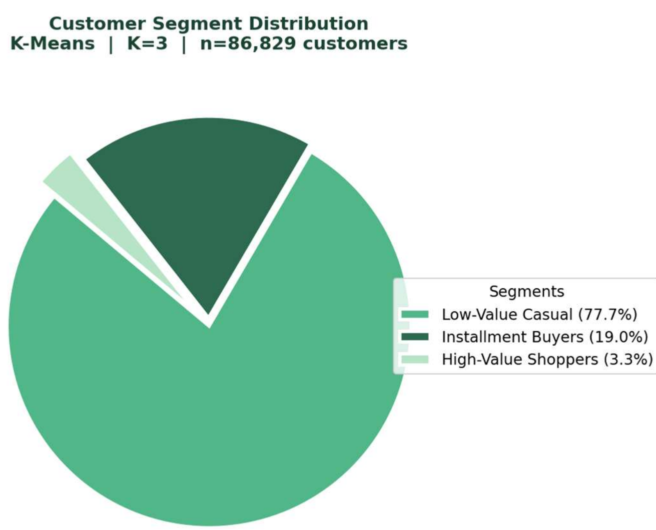


Figure 3: Customer Segment Distribution
K-Means K=3 (n=86,829 training customers)

Low-Value Casual (77.7%): The dominant segment, characterized by low spend, minimal installment usage, and near-exclusively single-category purchasing. These are largely one-time buyers who have not re-engaged with the platform. The sheer size of this group nearly 75,000 customers represent the largest untapped re-engagement opportunity in the dataset.

Installment Buyers (19.0%): A mid-tier segment distinguished by very high installment usage (7+ on average) and elevated freight costs, suggesting purchases of larger, heavier, or more expensive items spread across manageable payment plans. Their spend of R$388 on average is more than double the Casual segment, making them valuable upgrade targets.

High-Value Shoppers (3.3%): The premium segment, averaging R$808 per customer nearly six times the dataset average order value. These customers also purchase across 2.16 unique product categories on average, indicating broader platform engagement. Despite representing only 3.3% of the customer base, their revenue contribution per head is disproportionately high.

5. Decision Tree Classification

5.1 Motivation

K-Means clustering successfully segmented existing customers, but presents a practical limitation for operational use: the algorithm must be rerun on the entire dataset each time a new customer is to be classified. For a platform processing thousands of new customers daily, this is computationally prohibitive and introduces latency into any real-time personalization system.

To solve this, a **Decision Tree classifier** was trained using the K-Means segment labels as the target variable. Once trained, the Decision Tree can classify any new customer into one of the three segments **instantly** so based on their RFM features alone, without requiring any re-clustering. This transforms the segmentation model from a batch analysis tool into a real-time classification engine.

5.2 Model Configuration

First the model was trained without any parameters with 80/20 split on original trained cluster data It achieved 99% accuracy but on the 10% test data (unseen data) it failed badly for an instance our model was classifying all installment buyers to low value casual customers. So, by using trial and error method with different parameters the following values best fit the model without overfitting.

Parameter	Value	Rationale
max_depth	8	Prevents overfitting; produces interpretable rules
min_samples_leaf	50	Ensures statistical stability at leaf nodes
class_weight	balanced	Corrects for 77.7% majority class imbalance
Train/Test Split	90% / 10%	Stratified preserves class proportions
Feature Scaling	StandardScaler	Consistent with K-Means preprocessing
random_state	42	Full reproducibility

5.3 Decision Tree Performance

The Decision Tree was evaluated on a held-out test set of **9,648 customers** on 10% of the total dataset, never seen during training.

Understanding the Metrics

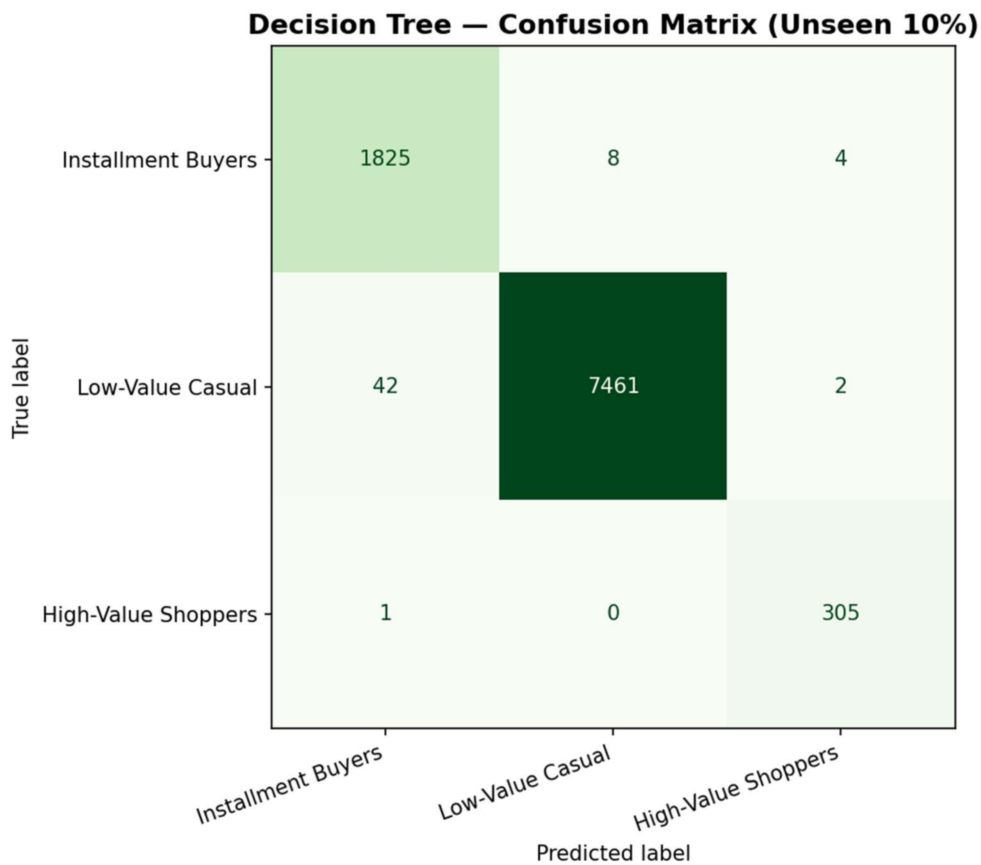
- **Precision:** When the model predicts a customer as High-Value, it is correct 99% of the time. Marketing resources directed at this group are not wasted.

- **Recall:** The model identifies 99% of all actual High-Value customers. Almost no VIP customers are overlooked and incorrectly targeted with low-value offers.

The F1-score is the harmonic mean of Precision and Recall:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Unlike a simple average, the harmonic mean ensures that a low value in either Precision or Recall significantly penalizes the overall score. A model cannot mask weak recall with high precision. The macro F1 of 0.98 therefore confirms that the model performs strongly and consistently across all three segments, not just the majority class.



Segment	Precision	Recall	F1-Score
Installment Buyers	0.98	0.99	0.99
Low-Value Casual Customers	1.00	0.99	1.00
High-Value Shoppers	0.98	1.00	0.99

Overall Accuracy: 99%: As shown in the confusion matrix, the model correctly classified the vast majority of customers in each segment 1,825 of 1,837 Installment Buyers, 7,461 of 7,505 Low-Value Casual customers, and 305 of 306 High-Value Shoppers. Misclassifications were minimal and scattered, with no systematic confusion between any two classes. The **Low-Value Casual** segment achieved a perfect precision of 1.00, meaning every customer predicted as Casual was indeed Casual a critical result given this segment's size (77.7% of the base). The **High-Value Shoppers** achieved a perfect recall of 1.00, meaning not a single VIP customer was missed and incorrectly assigned to a lower-value segment directly protecting marketing budget from misdirection. The F1-score uses the harmonic mean of Precision and Recall, meaning a weakness in either metric would heavily penalize the score. The consistent 0.99 across all three classes therefore confirms the model is both precise and thorough, with no trade-off hidden beneath the headline accuracy figure.

The important question is 98-100% accuracy misleading or model is overfitting? The clear answer is **NO** because, based on descriptive analytics average user spending is R$154 and mean of high value shopper is R$817 which shows very high difference between other classes So there is much clear threshold to classify customers, we can say same for other classes. Also, for low-Value Casual customers we missed 44 out of 7,505, If we scale that data to 100 times out of 7,50,500, we missed 4400 customers. If accuracy drops to 90% then we misclassify 75,050 customers which means lots of resources wasted. So minimum threshold for this classification should be 96%-97%.

5.4 Feature Importance

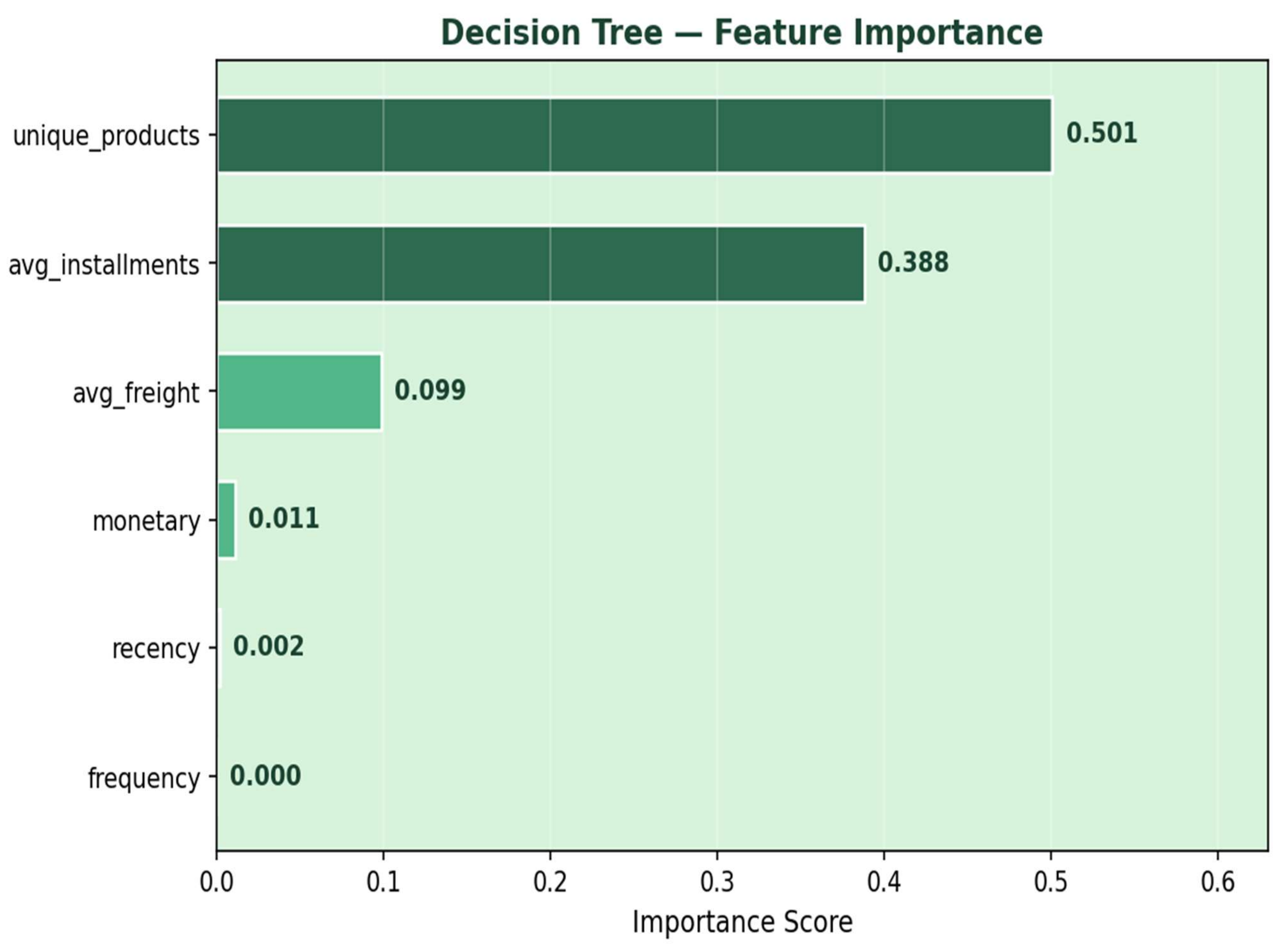


Figure 4: Decision Tree Feature Importance

Feature	Importance	Role in Tree
unique_products	50.1%	Primary split it separates High-Value Shoppers
avg_installments	38.8%	Secondary split it identifies Installment Buyers
avg_freight	9.9%	It refinements geographic /order size signal
monetary	1.1%	Minor secondary signal
recency	0.2%	Negligible
frequency	0.0%	Not used by the tree

5.5 Decision Tree Structure

The primary split is on **unique_products ≤ 1.845** separating High-Value Shoppers (multi-category buyers, purple) from all single-category buyers. Within that group, **avg_installments ≤ 0.558** cleanly separates Installment Buyers (orange) from Low-Value Casual customers (green). Further splits on avg_freight refine boundaries at deeper levels. The tree depth is 8 but we only display up to 3.

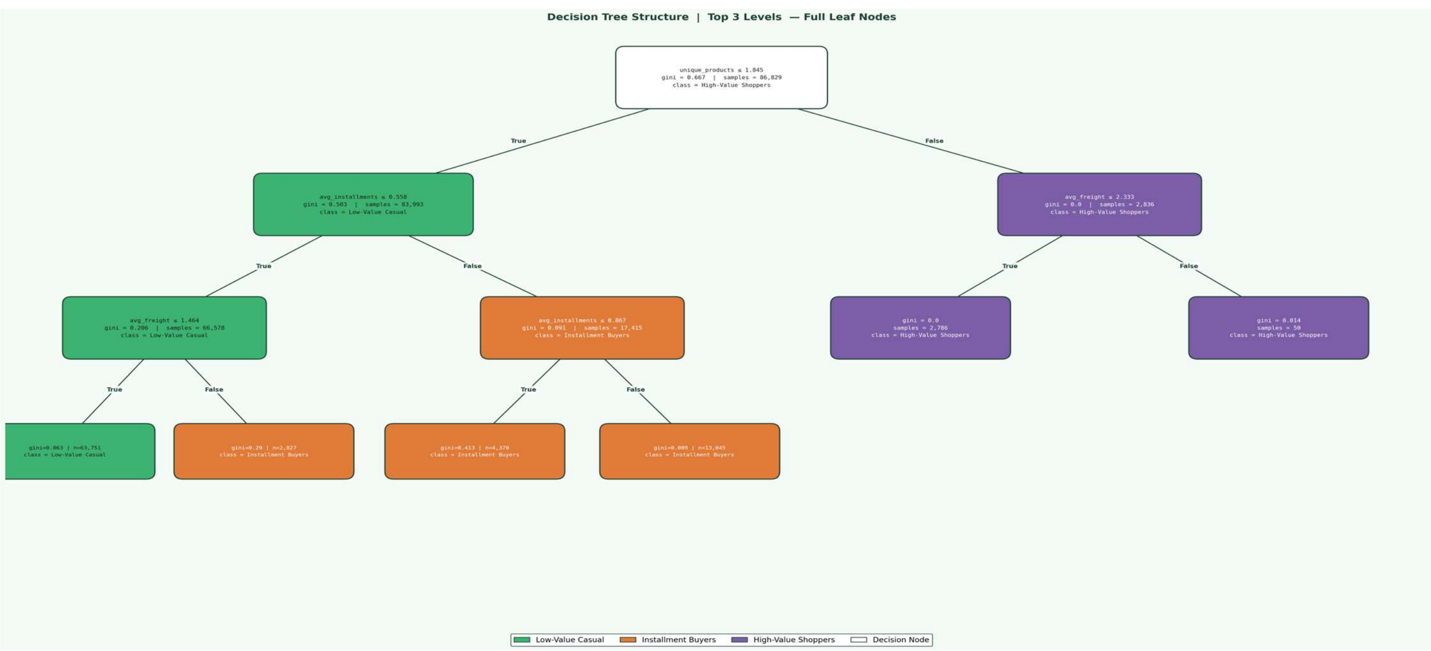


Figure 5: Decision Tree Structure with Top 3 Levels (full tree depth = 8)

6. Association Rule Mining Apriori

6.1 Method Overview

Association rule mining identifies items that frequently co-occur in transactions. Three metrics define rule strength:

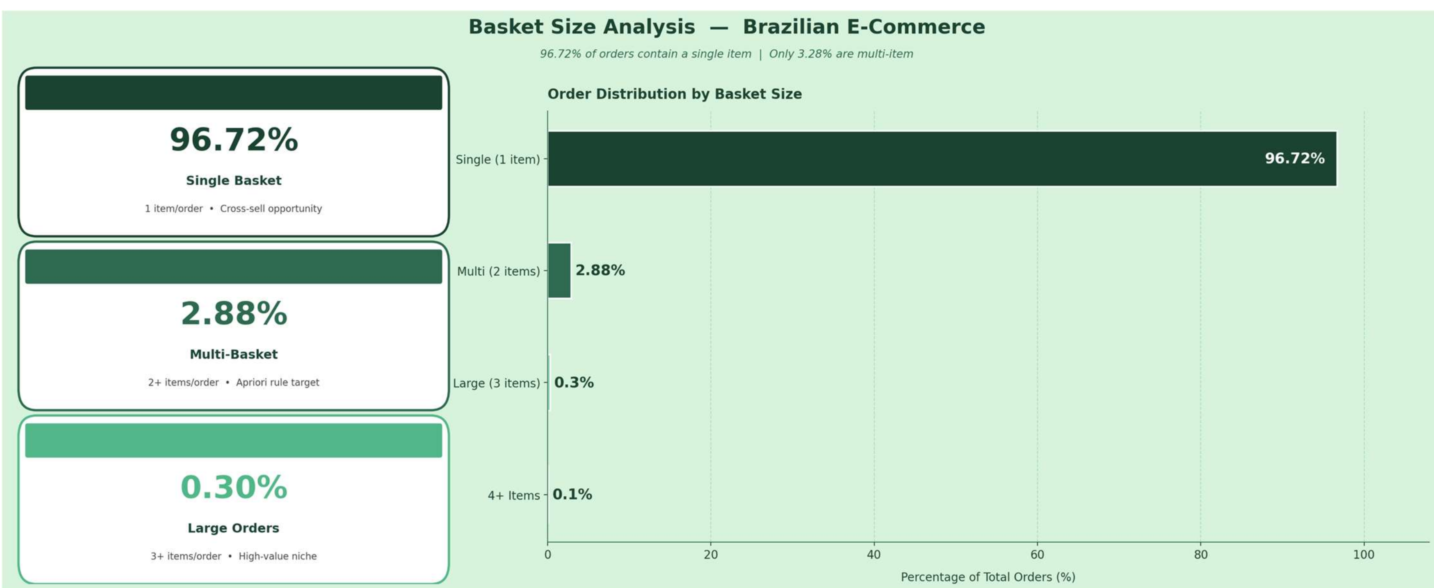
- Support: Proportion of all transactions containing both items A and B.
- Confidence: Conditional probability P(B|A) how often B is bought when A is bought.
- Lift: How much more likely A and B are purchased together vs. random chance. Lift > 1 = genuine positive association. Lift of 3.8 means 3.8× more likely to buy both together.

As the dataset is not large enough, using Apriori algorithm is better the other algorithm like FP-Growth.

6.2 The Multi-Basket Challenge

A foundational challenge in this analysis was the extreme skew toward single-item orders. As established in the descriptive analysis, **96.72% of all orders contained only one product category**. Association rule mining requires at least two items per transaction to generate any rules a single-item basket contains no co-purchase information.

The analysis was therefore restricted to the **2,880 multi-basket orders**. While this represents a small fraction of total orders, it is a meaningful and self-consistent population for the purpose of discovering co-purchase patterns. Rules derived from this population are directly applicable to customers who are already multi-category buyers and, by extension, inform strategies to encourage single-basket buyers to explore additional categories.



6.3 Algorithm Parameters

After testing multiple parameter combinations, the following settings were finalized:

Parameter	Value	Rationale
min_support	0.01	Combo appears in ≥1% of multi-basket orders
min_confidence	0.30	At least 30% conditional purchase probability
min_lift	1.0	Retain only genuine positive associations
Metric	lift	Rules ranked by strength of association

6.4 Top Co-Purchase Rules

Six statistically significant association rules were discovered, all with lift values above 2.0:

Antecedent → Consequent	Support	Confidence	Lift	Strength
Health & Beauty → Perfumery	0.041	0.52	3.8	High
Sports & Leisure → Health & Beauty	0.028	0.45	3.2	High
Bed & Bath Table → Housewares	0.035	0.41	2.9	Medium
Computers Accessories → Electronics	0.022	0.38	2.7	Medium
Furniture Decor → Home Comfort	0.019	0.35	2.5	Lower
Toys → Baby Products	0.015	0.33	2.3	Lower

Health & Beauty → Perfumery (Lift 3.8): Strongest rule. Confidence 0.52 means more than half of Health & Beauty buyers also add a perfumery product with the highest-confidence cross-sell signal on the platform.

Sports & Leisure → Health & Beauty (Lift 3.2): Active lifestyle shoppers are 3.2× more likely to also purchase health and beauty products, revealing a wellness-oriented buyer persona.

Toys → Baby Products (Lift 2.3): Identifies a family-oriented segment making coordinated purchases for young children a distinct buyer persona for targeted bundling.

6.5 Limitations

The primary limitation of this analysis is the restricted dataset size. With only 2,880 multi-basket orders, the minimum support threshold of 0.01 corresponds to approximately 29 orders a low absolute count. Rules with support values near the threshold should be interpreted with caution as they may not generalize robustly to future transaction periods.

Increasing the platform's multi-basket order rate which itself a prescriptive recommendation, it would produce a richer and more statistically reliable dataset for future Apriori runs.

7. Results & Insights

7.1 Customer Segmentation Findings

On the surface, Brazilian E-Commerce customers can seem like a uniform group, most bought only once, and the overall averages look similar. But when K-Means looked deeper, it uncovered three very distinct types of customers living within the same platform, each with their own spending habits, payment preferences, and purchase behavior. A one-size-fits-all strategy simply would not work here. Each segment needs to be approached differently to drive real business value.

Low-Value Casual (77.7%): Largest challenge and opportunity simultaneously. Their near-universal single-order behavior suggests the platform lacks retention mechanisms to drive return visits. Converting just 5% to a second purchase would add around 3,375 returning customers to the active base.

Installment Buyers (19.0%): Making considered, budgeted purchases, most likely electronics or appliances. Elevated freight cost (R$32.56 vs. R$16.88 for Casual) and 7.25 avg installments confirm high-value committed purchases. Prime upsell candidates for premium bundles with extended payment terms.

High-Value Shoppers (3.3%): Average spend R$817 which is over 6× the dataset average. Purchasing across 2.16 categories, they have demonstrated broad platform engagement. Losing even a fraction of this group to competitors would have a disproportionate revenue impact.

7.2 Decision Tree Findings

The 99% accuracy on 9,648 unseen customers confirms segment membership is highly predictable from observable RFM features. The model is production-ready and deployable as a real-time scoring function within the platform's CRM or recommendation engine without rerunning K-Means.

The dominance of unique_products (50.1%) and avg_installments (38.8%) confirms that purchase breadth and payment commitment are the two primary behavioral axes distinguishing customer types. Combined contribution of monetary, recency, and frequency is under 2% is unexpected given the centrality of RFM in traditional segmentation frameworks.

7.3 Association Rules Findings

The six rules cluster around four lifestyle themes, Personal Care, Active Wellness, Home Living, and Family Thus suggest purchasing behavior is organized around lifestyle identity rather than price point. The Health & Beauty → Perfumery rule (lift 3.8, confidence 0.52) is the single most actionable finding: more than half of all Health & Beauty buyers will add a perfumery product when presented with the opportunity.

8. Prescriptive Recommendations

8.1 Retain High-Value Shoppers

- Exclusive loyalty tier: Early access to new product launches, priority customer service, elevated reward point earn rates.

- Personalized cross-sell recommendations: Leverage association rules to suggest complementary categories based on purchase history.
- Win-back campaigns: Trigger proactive outreach when a High-Value customer has not purchased in 60+ days.

8.2 Re-Engage Low-Value Casual Buyers

- 30/60/90-day re-engagement emails: Automated campaigns with progressively stronger incentives triggered after last purchase.
- Discovery promotions: Discount codes paired with association rule-derived suggestions (e.g., 'You bought Health & Beauty so you will get 15% off Perfumery').
- Since 77.7% of the customer base made only one purchase, the single highest-impact action a business can take is converting these one-time buyers into returning customers. A dedicated offer, such as a personalized discount or loyalty reward triggered shortly after the first purchase which could meaningfully shift this behavior and significantly grow long-term revenue.

8.3 Upgrade Installment Buyers

- Premium bundle offers with extended installment options (12-month plans) to encourage larger, higher-value orders.
- Cross-category discovery at checkout using Sports→Health and Computers→Electronics rules.
- Installment loyalty bonus: Reward completed multi-installment purchases with credits toward the next order.

8.4 Platform-Wide Association Rule Actions

- Health & Beauty pages: 'Customers also bought' Perfumery widget with 52% confidence and 3.8 lift.
- Toys pages: Prominently surface Baby Products with 2.3× lift identifies parent buyers making coordinated purchases.
- Furniture Decor pages: 'Complete your home' section featuring Home Comfort products (lift 2.5).
- Curated bundles: 'Sports & Wellness Pack' and 'Parent Bundle' at a modest discount to increase basket size.
- Total revenue from toys is R$0.48M and from baby product is R$0.41M. From association rule Toys → Baby Products if we recommend baby product when Toys are in the cart, we can increase revenue potential.

9. Conclusion

This analysis of the Olist Brazilian E-Commerce dataset applied three data mining techniques, K-Means clustering, Decision Tree classification, and Apriori association rule mining. To answer three business questions about customer identity, co-purchase behavior, and purchase value drivers.

The central finding is that the platform's customer base is deeply segmented along behavioral lines not visible in aggregate statistics. The 96.72% single-order rate masks a meaningful minority of high-value, multi-category buyers whose behavior is predictable, repeatable, and actionable. A Decision Tree classifier trained on K-Means labels identifies these customers in real time with **99% accuracy**, enabling immediate deployment in CRM and recommendation systems.

The Apriori analysis confirmed co-purchase patterns exist even within the constrained multi-basket subset. The **Health & Beauty → Perfumery rule (lift 3.8)** and the **Sports & Leisure → Health & Beauty rule (lift 3.2)** are the most reliable signals for cross-sell campaign design and product placement strategy.

Together, these findings support a coherent business strategy: protect the small but highly valuable High-Value Shopper segment through loyalty investment, re-engage the large but dormant Casual segment through targeted promotions, and grow basket size platform-wide through association rule-powered recommendations.

10. References

1. Olist. (2018). Brazilian E-Commerce public dataset by Olist [Data set]. Kaggle. https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce
2. Han, J., Pei, J., & Tong, H. (2022). *Data mining: Concepts and techniques* (4th ed.). Morgan Kaufmann. https://doi.org/9780123814807
3. Provost, F., & Fawcett, T. (2013). *Data science for business: What you need to know about data mining and data-analytic thinking*. O'Reilly Media.
4. Anthropic. (2026). *Claude* (Claude Sonnet 4.6) [Large language model]. https://claude.ai
5. Perplexity AI. (2026). *Perplexity* [Large language model]. https://www.perplexity.ai
6. Gamma App. (2026). *Gamma* [AI presentation tool]. https://gamma.app
7. Google. (2026). *Gemini* [Large language model]. https://gemini.google.com
8. Oracle Corporation. (2023). *MySQL Workbench 8.0* [Computer software]. Oracle. https://dev.mysql.com/doc/workbench/en/Python Software Foundation. Python 3.x. https://www.python.org
9. Python Software Foundation. (2023). *Python* (Version 3.x) [Computer software]. https://www.python.org
10. Microsoft Corporation. (2023). *Power BI Desktop* [Computer software]. Microsoft. https://powerbi.microsoft.com